A 16-bit 1-MS/s SAR ADC With Capacitor Mismatch Self-Calibration

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***Abstract*— This article introduces a successive approximation register (SAR) analog-to-digital converter (ADC) that utilizes a foreground capacitor mismatch self-calibration method. The proposed floating operation puts the uncalibrated high-bit capacitor into the floating state, preventing the sub-ADC from saturating caused by comparator static offset during the calibration process. To address the random mismatch of the LSB capacitors and improve the calibration accuracy, this article employs round-robin grouping of eight sets of LSB capacitors. In addition, a precharged bootstrapped switch is proposed to achieve high sampling linearity with low power consumption and area overhead. An anti-interference custom-designed 0.5-fF capacitor structure is suggested for binary-weighted capacitor mismatch of capacitive DAC (CDAC). Furthermore, the circuit implementation of the comparator utilized by ADC is also discussed. The prototype was fabricated in a 180-nm CMOS process with a 1.8-V supply and achieved spurious-free dynamic ranges of 108.9 and 92.38 dB at an input frequency of 1 kHz while operating at sampling rates of 100 kS/s and 1 MS/s, respectively. The prototype consumes 6.745 mW and occupies 0.91 mm^2.**

***Index Terms*— Bootstrapped switch, capacitor mismatch, custom-designed capacitor, foreground calibration, self-calibration, successive approximation register (SAR) analog-to-digital converter (ADC).**

I. Introduction

SUCCESSIVE approximation register (SAR) analog-to-digital converters (ADCs) have gained considerable research interest and widespread application with the superior power efficiency and excellent compatibility with the advanced CMOS technology. Due to its simple architecture and lack of requirement for op-amps, the SAR ADC is also an excellent option for hybrid-architecture ADCs, such as flash-SAR, pipeline-SAR, and noise-shaping-SAR. Unlike other architectures that are limited to relatively fixed application scenarios, SAR ADCs offer a broader range of applications, particularly for medium-to-high speed and resolution. In the applications of digital imaging, industrial control, and high-precision instrumentation, where high precision (16 bit or above) and moderate speed (below a few megasamples per seconds) are required, the SAR ADC is better suited than the sigma–delta ADC. In contrast to sigma–delta ADCs, SAR ADCs achieve one-time sampling and conversion without the need for signal postprocessing, thereby enhancing their capability to handle multiplexed inputs.

The linearity of SAR ADC is primarily constrained by the capacitor mismatch of capacitive DAC (CDAC), which imposes a limit of 12 bits on the resolution of conventional SAR ADC [1], [2]. Two main factors need to be considered before designing a CDAC: the kT/C noise and capacitor mismatch. In order to achieve the necessary linearity, the size of the CDAC must be significantly increased compared with when only considering kT/C noise, as smaller capacitors are more susceptible to mismatch. The larger size of the CDAC not only leads to increased area and power consumption but also adds to the burden on the front-end module. In [3], [4], [5], [6], [7], and [8], pipelined SAR architecture was adopted to realize high linearity, including a primary SAR ADC, a secondary SAR ADC, and a residue amplifier. Although the linearity requirement for each stage of SAR ADC can be mitigated by pipelining, it transfers the design difficulty to the residue amplifier. Laser trimming can provide necessary capacitor matching at a high cost, but trimmed capacitance may still be mismatched due to package stress or long-term drift. The capacitor calibration presents an alternative approach to enhance CDAC linearity by detecting and compensating for mismatch errors, which can be classified into background and foreground calibration. The background calibration is performed simultaneously with the ADC normal conversion. [9], [10] detect the capacitor mismatch error of CDAC by injecting discrete-time pseudorandom noise and then extracting the bit weight by least-mean-square (LMS) adaptive loop. McNeill et al. [11] split a single ADC into two independent halves and performs two SAR conversions simultaneously. The difference between the two outputs of half-sized ADCs is used to estimate the capacitor mismatch through an LMS loop. The foreground calibration technique detects the mismatch prior to the normal conversion of the ADC. Kerzérho et al. [12], Adamo et al. [13], and Handel et al. [14] perform the calibration by adjusting the bit weight to minimize the error between the reconstructed output and the known input. However, these techniques involve the processing of extensive data and have input signal requirements, which can make calibration more complex. Self-calibration is performed only by the intrinsic block of SAR ADC and does not require input,

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which features the minimum hardware overhead and is not subject to convergence issues [15], [16], [17], [18], [19], [20], [21], [22]. Mismatch errors can be compensated in the analog domain using an auxiliary DAC [22] or in the digital domain by reconstructing the bit weights [15], [16], [17], [18], [19], [20], [21].

Utilizing the self-calibration method based on bottom–up, the high-bit capacitors of the CDAC are first quantized by the sub-ADC consisting of low-bit capacitors, comparator, and SAR logic. Subsequently, the weights of the calibrated capacitor are reconstructed in the digital domain using quantization results and employed during ADC normal operation. However, the precision of self-calibration will be decreased due to calibration errors caused by static offset of the comparator and mismatched LSB capacitors. In [18] and [19], additional logic is incorporated to control the capacitor array for compensating the static offset prior to capacitor calibration, which increases hardware overhead and calibration complexity. In this article, we propose a floating operation that overcomes offset problems without introducing extra logic overhead. Moreover, combined with a custom-designed capacitor, round-robin for eight groups of LSB capacitors is taken for each bit calibration to address the calibration error related to LSB capacitors mismatch.

In this article, we present a 16-bit binary-weighted SAR ADC implemented in 180-nm CMOS technology. The theory of the capacitor mismatch calibration technique employed in the ADC was initially introduced in [21]. This article presents a more comprehensive analysis of the calibration technique, taking into account the impact of parasitic capacitance and noise. Also, the practicality and effectiveness of this technique are further demonstrated through on-chip testing. Furthermore, it also explores other high-precision techniques utilized for this ADC. First, we propose a precharged bootstrapped switch to enhance the linearity of ADC sampling, thereby reducing both area and power overhead compared with conventional structures. Second, instead of employing a split CDAC structure with bridge capacitors, we adopt a binary CDAC based on a custom-designed anti-interference 0.5-fF capacitor to avoid gain errors between the split CDACs. In addition, the architecture and circuit implementation of the comparator utilized in the ADC are discussed in this article.

This article is organized as follows. Section II introduces a foreground capacitor mismatch self-calibration technique, which is validated through the behavioral model simulations. In Section III, implementation details of the circuit are presented, including a precharged bootstrapped switch, custom-designed capacitor, and comparator. Measurement results of the prototype chip are shown in Section IV. Finally, Section V concludes this article.

II. High-Precision Capacitor Mismatch Self-Calibration Technique

A. Theory and Limitations of Self-Calibration Method Based on Bottom–Up

Fig. 1 illustrates a self-calibration method based on bottom–up for capacitor mismatch, the capacitor under

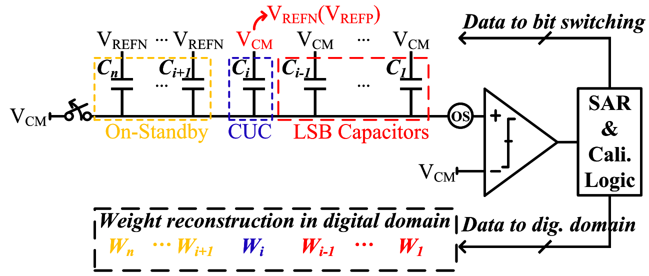

Fig. 1. Self-calibration based on bottom–up using LSB capacitors.

calibration (CUC) C_i is calibrated using the LSB capacitors which is considered to have a desirable binary weighting. The calibration process can be divided into three phases.

Reset: The comparator inputs are reset to V_{CM}, the CUC C_i and the LSB capacitors are set to V_{CM} while the higher bits hold at V_{REFN}.

Reverse: The comparator input net is disconnected from V_{CM} first to prevent charge injection errors. Then, the SAR and calibration block forces C_i switch from V_{CM} to V_{REFN}, inducing a negative voltage step on the comparator input

$$\Delta V_{i,\mathrm{force0}} = \frac{C_i}{C_{\mathrm{tot}}}(V_{\mathrm{REFN}} - V_{\mathrm{CM}}). \tag{1}$$

Quantization: A sub-ADC composed of the inherent comparator, SAR logic, and LSB capacitors quantize the negative voltage step related by C_i and comparator offset

$$-\frac{C_i}{C_{\mathrm{tot}}} \cdot V_{\mathrm{CM}} + V_{\mathrm{os}} = \sum_{j=1}^{i-1} \frac{D_j C_j}{C_{\mathrm{tot}}} \cdot V_{\mathrm{CM}}. \tag{2}$$

Meanwhile, the corresponding weight calculation of (2) in digital domain can be expressed as follows:

$$W_{i,\mathrm{force0}} = \sum_{j=1}^{i-1} D_j W_j. \tag{3}$$

Then, a similar process is performed but the C_i switches from V_{CM} to V_{REFP}, causing a positive voltage variation which is quantized by the sub-ADC and calculated in the digital domain

$$\frac{C_i}{C_{\mathrm{tot}}} \cdot V_{\mathrm{CM}} + V_{\mathrm{os}} = \sum_{k=1}^{i-1} \frac{D_k C_k}{C_{\mathrm{tot}}} \cdot V_{\mathrm{CM}} \tag{4}$$

$$W_{i,\mathrm{force1}} = \sum_{k=1}^{i-1} D_k W_k. \tag{5}$$

By taking the difference between (2) and (4), the static offset can be eliminated, the calibrated weight of C_i can be expressed as follows:

$$W_{i,\mathrm{cal}} = \left(\sum_{k=1}^{i-1} D_k W_k - \sum_{j=1}^{i-1} D_j W_j \right) / 2. \tag{6}$$

The calibration process will be repeated multiple times, and the final reconstruction weight of C_i will be determined as the average of the multiple calibration values, thereby mitigating the thermal noise errors. The calibrated C_i will be used to measure the next capacitor C_{i+1}, the process continues until all capacitors have been calibrated.

According to (2), (4), and (6), the static offset will not affect the calibration accuracy of CUC. However, if the sum of the static offset and voltage variation caused by CUC is outside the quantization range of the sub-ADC, then (6) leads to an incorrect calibration value. The tolerable range for static offset under calibration of C_i can be expressed as follows:

$$\frac{C_i - \sum_{j=1}^{i-1} C_j}{C_{\text{tot}}} \cdot V_{\text{CM}} < V_{\text{os}} < \frac{\sum_{j=1}^{i-1} C_j - C_i}{C_{\text{tot}}} \cdot V_{\text{CM}}. \quad (7)$$

From (7), the utilization of redundant bits can expand the acceptable range for offset, but a large number of redundancy bits will limit the conversion speed of the ADC during normal operation. In [18] and [19], extra capacitor arrays are used to correct the static offsets before the calibration for CUC, resulting in additional hardware overhead and logic complexity.

Self-calibration from bottom to up has a basic principle that the LSB group needs to have a precise weight so that it need not be calibrated. In Fig. 1, the calibrated value of C_i storing in the digital domain can be expressed as follows:

$$C_{i,\text{cal}} = \sum_{j=1}^{i-1} D_j C_{j,\text{nom}} \quad (8)$$

where $C_{j,\text{nom}}$ represents the nominal capacitor of C_j. In fact, the random mismatch of LSB capacitors is not negligible. A calibration error exists between C_i and $C_{i,\text{cal}}$ due to the LSB mismatch

$$C_j = C_{j,\text{nom}} + e_{j,\text{mis}}$$
$$e_{i,\text{cal}} = C_i - C_{i,\text{cal}} = \sum_{j=1}^{i-1} D_j e_{j,\text{mis}}. \quad (9)$$

Unlike random noise, calibration error caused by mismatch of LSB capacitors cannot be eliminated via averaging, and the calibration error of the low-bit capacitor also accumulates exponentially with the calibration proceeding, thus limiting the calibration precision. One solution is to increase the area of the LSB capacitors since a k-fold increase in the area of the capacitor changes its mismatch factor to $1/\sqrt{k}$-fold, but this likewise makes the area of the CDAC increase by a factor of k as well.

B. Proposed Calibration Techniques

This article presents a method with less overhead using floating capacitors to increase the tolerance range for offsets of the sub-ADC, as shown in Fig. 2.

Unlike in Fig. 1, the uncalibrated capacitor is set to float under the calibration process of CUC. Note that the proportion of parasitic capacitance C_P, resulting from the input MOS of comparator and wires routing, to the overall capacitance of the top plate of CDAC increases due to the floating of the high-bit capacitors. Although C_P will change slightly because the state of CDAC is different for each calibrated capacitor bit, it does not introduce calibration error because the voltage variation generated by the switching of C_i and the sub-ADC attenuates

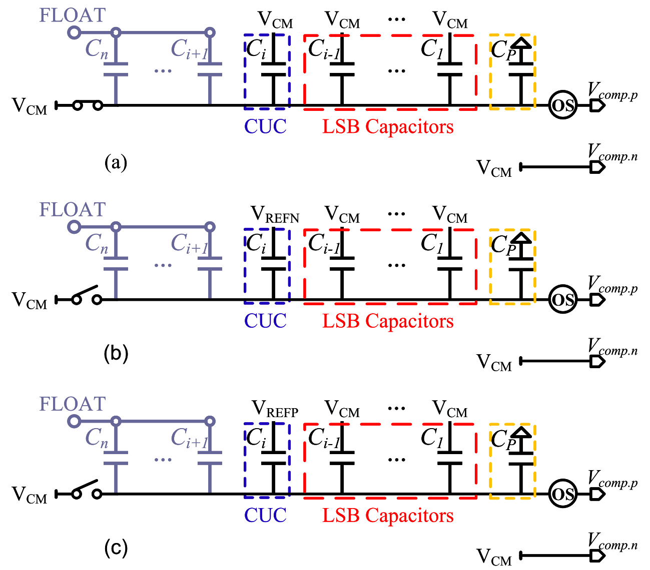

Fig. 2. Calibration with floating capacitors operation. (a) Reset phase. (b) Switch CUC from V_{CM} to V_{REFN}. (c) Switch CUC from V_{CM} to V_{REFP}.

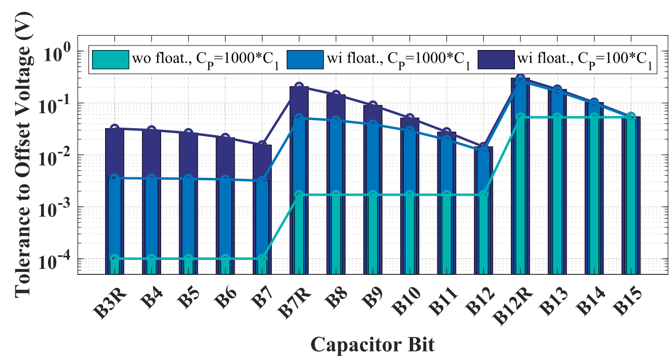

Fig. 3. Tolerance to offset before/after floating capacitors operation.

with the same coefficient according to (2) and (4). Thus, the acceptable range of static offset can be expressed as follows:

$$\frac{C_i - \sum_{j=1}^{i-1} C_j}{\sum_{j=1}^{i} C_j + C_P} \cdot V_{\text{CM}} < V_{\text{os}} < \frac{\sum_{j=1}^{i-1} C_j - C_i}{\sum_{j=1}^{i} C_j + C_P} \cdot V_{\text{CM}}. \quad (10)$$

Fig. 3 illustrates the tolerance to the static offset voltage when each bit of the capacitor is calibrated before and after the floating operation using B3–B0 as the LSB capacitors in this design, where C_P is set to $100 * C_1$ and $1000 * C_1$, respectively. The CDAC inserts three radix-1 redundant bits, B3R, B7R, and B12R, which not only corrects the comparator decision error due to settling but also ensures that the quantization range of the sub-ADC covers the CUC mismatch and the static offset. As depicted in Fig. 3, when considering only the effect of redundant bits, the calibration of B12R and B13–B15 experiences a relaxed offset requirement as they benefit from all the redundant bits; whereas, the low-bits capacitors are calibrated with a stringent condition, where the calibration of B7R and B8–B12 allows an offset voltage within ± 1.7 mV. The challenge lies in the fact that the calibration of B3R and B4–B7 only allows for an offset voltage within ± 100 μV, which poses a difficulty for the comparator to achieve, even with the autozero technique. The floating capacitor operation effectively addresses this issue. According to Fig. 3, when calibrating the B7R and higher bits capacitors under the same C_P (1000 * C_1), the offset tolerance range exceeds ±11.9 mV due to floating operation. Also, capacitor calibration of B3R and B4–B7 experience the more relaxed offset requirements of ±3.6 mV than the case without floating. Although this operation cannot be applied in calibrating B15 since no capacitor can be floated, it still benefits from the preceding redundant bits, which allow for an offset tolerance of ±53 mV.

As shown in Fig. 3, C_P reduces the effect of floating operation. However, when compared with the nonfloating case, employing floating capacitors significantly enhances the acceptable range of offset for sub-ADC. Consequently, it effectively avoids calibration errors caused by sub-ADC saturation due to static comparator offset, particularly when combined with the autozero technique integrated into the comparator.

Although averaging does not resolve the issue of offset, it can effectively attenuate the noise during the calibration process. According to Fig. 2, when calibrating the C_i, the LSB voltage of the $(i-1)$-bit sub-ADC can be expressed as follows:

$$V_{\text{LSB},i} = \frac{C_1}{\sum_{j=1}^{i} C_j + C_P} \cdot V_{\text{REFP}}. \tag{11}$$

Also, the rms value of the sampling thermal noise can be expressed as follows:

$$V_{n,\text{thr},i} = \sqrt{\frac{kT}{\sum_{j=1}^{i} C_j + C_P}}. \tag{12}$$

The quantization noise is denoted as follows:

$$V_{n,\text{quan},i} = \frac{V_{\text{LSB},i}}{\sqrt{12}}. \tag{13}$$

We use $V_{n,\text{comp}}$ to represent the input-referred noise of the comparator, so the total noise at the input of the comparator when calibrating the ith bit capacitor can be expressed as follows:

$$V_{n,\text{tot},i} = \sqrt{V_{n,\text{thr},i}^2 + V_{n,\text{quan},i}^2 + V_{n,\text{comp}}^2}. \tag{14}$$

Because the noise is reduced by the averaging operation, to evaluate the impact of noise on the quantization of sub-ADC during calibration, we introduce α_i as the ratio between the LSB voltage and the rms noise of the sub-ADC when calibrating C_i

$$\alpha_i = \frac{V_{\text{LSB},i}}{V_{n,\text{tot,avg},i}}, \quad V_{n,\text{tot,avg},i} = \frac{V_{n,\text{tot},i}}{\sqrt{N_{\text{avg}}}} \tag{15}$$

with C_1 set to 0.5 fF, C_P set to 1000 * C_1, V_{REFP} set to 1.8 V, $V_{n,\text{comp}}$ set to 15 μV, and N_{avg} denotes the average times when calibrating each bit capacitor.

Fig. 4 illustrates the value of α when calibrating each bit capacitor for different average times. α without floating operation is the same for all capacitor bits because V_{LSB} and $V_{n,\text{tot}}$ are constant during calibration. For the calibration of lower bits capacitors below B15, the proposed floating operation has better noise performance with higher α. The worst α with floating operation occurs when calibrating B15, which is the same as the case without floating. Therefore, the floating operation does not introduce more averaging times and hardware overhead due to noise. The value of α decreases as the calibration bits increase due to a faster decrease in V_{LSB} compared with the $V_{n,\text{thr}}$, as indicated in (11) and (12). When there is no averaging, according to Fig. 4, α_{B3R} and α_{B15} are equal to 3.4 and 1.9, respectively, indicating that the total rms noise voltage is lower than the LSB voltage of the sub-ADC for each bit calibration with the floating operation. However, in order to completely eliminate the impact of noise, averaging is necessary. Generally, for each bit calibration, the relation $3 * V_{n,\text{tot,avg},i} < 0.5 * V_{\text{LSB},i}$ needs to be guaranteed. According to Fig. 4, it is satisfied when the average times is taken as 16 or more.

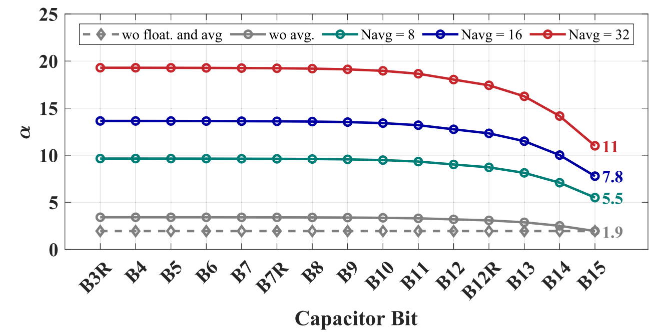

Fig. 4. Value of α for different average times.

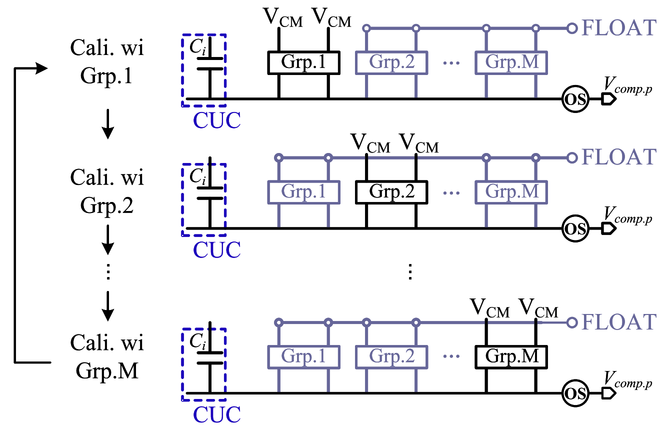

Fig. 5. Calibration using multiple groups of LSB capacitors.

In conjunction with the averaging operation, the method using multiple groups of LSB capacitors round-robin is utilized to reduce calibration errors caused by mismatched LSB capacitors, as shown in Fig. 5. During N times repeated calibrations of C_i, one of M groups of LSB capacitors is enabled in turn while other unused groups hold floating. We use $C_{i,\text{Tk}}$ to represent the kth quantization value of C_i, and $C_{j,\text{Gm}}$ to represent the capacitor in the mth group LSB capacitors. Moreover, we assume that each of the repeated calibrations for C_i has the same digital output for the sake of simplicity. The quantization values corresponding to M groups of LSB capacitors are

$$\begin{cases} C_{i,T1} = \sum_{j=1}^{i-1} D_j C_{j,G1} \\ C_{i,T2} = \sum_{j=1}^{i-1} D_j C_{j,G2} \\ \cdots \\ C_{i,\mathrm{TM}} = \sum_{j=1}^{i-1} D_j C_{j,\mathrm{GM}}. \end{cases} \tag{16}$$

According to (16), the calibration error of C_i for each quantization are

$$\begin{cases} e_{i,\mathrm{cal},T1} = \sum_{j=1}^{i-1} D_j e_{j,\mathrm{mis},G1} \\ e_{i,\mathrm{cal},T1} = \sum_{j=1}^{i-1} D_j e_{j,\mathrm{mis},G2} \\ \cdots \\ e_{i,\mathrm{cal},\mathrm{TM}} = \sum_{j=1}^{i-1} D_j e_{j,\mathrm{mis},\mathrm{GM}}. \end{cases} \tag{17}$$

When the calibration values of C_i are averaged N times in the digital domain to remove noise, the calibration error shown in (17) is averaged M times as long as N is larger than M

$$\begin{aligned} e_{i,\mathrm{cal,avg}} &= \frac{1}{M}\sum_{j=1}^{i-1} D_j \left(e_{j,\mathrm{mis},G1} + e_{j,\mathrm{mis},G2} + \cdots + e_{j,\mathrm{mis},\mathrm{GM}}\right) \\ &= \sum_{j=1}^{i-1} D_j e_{j,\mathrm{mis,avg}}. \end{aligned} \tag{18}$$

As the random mismatch error of capacitors follows Gaussian distribution

$$e_{j,\mathrm{mis}} \sim N\left(0, \sigma_{\mathrm{mis}}^2\right) \tag{19}$$

where the standard deviation σ_{mis} is determined by the manufacturing process, after averaging M times

$$e_{j,\mathrm{mis,avg}} \sim N\left(0, \frac{\sigma_{\mathrm{mis}}^2}{M}\right) \tag{20}$$

which means that the equivalent calibration error for C_i is minimized after averaging from the perspective of probability, thus reducing the impact of error accumulation. Note that the LSB capacitors have a small value, so the area overhead is negligible. Since the linearity of the CDAC is primarily determined by the matching of the high-bit capacitors, one group of LSB capacitors is used in the ADC normal operation mode to avoid excessive logic overhead.

Behavioral simulation of the proposed ADC is used to validate the effectiveness of the proposed calibration method. The behavioral model considers the CDAC with the same capacitance and random mismatch as the actual circuit, and the lower 4 bits of the CDAC are utilized for calibrating the higher bits capacitance. Each bit is averaged 32 times to ensure that the impact of noise and mismatch LSB capacitors is minimized. Fig. 6 illustrates the spectrum of different calibration schemes under identical CDAC mismatch and a static offset of 30 * LSB (1.65 mV) for comparator. Comparing performance, it is observed that using conventional self-calibration leads to degraded results (9.2-bits ENOB and 63.3-dB SFDR) compared with no calibration (11.1-bits ENOB and 74.5-dB SFDR). This degradation occurs due to miscalibration in low bits caused by static offset, resulting in cumulative errors during the calibration process. This issue is addressed by the proposed floating capacitor operation. By using the floating capacitor operation without LSB capacitors group round-robin, the ADC achieves an ENOB of 14.1 bits and an SFDR of 93.7 dB. With further improvement through calibration using eight groups of LSB capacitors round-robin technique, the ADC demonstrates enhanced performance with an ENOB of 15.5 bits and an SFDR value reaching up to 115.5 dB.

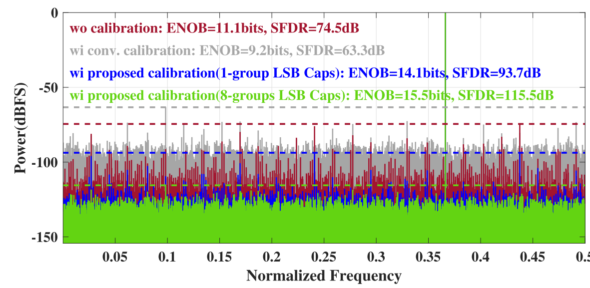

Fig. 6. Spectrum with different calibration methods under same mismatch and 30 * LSB offset.

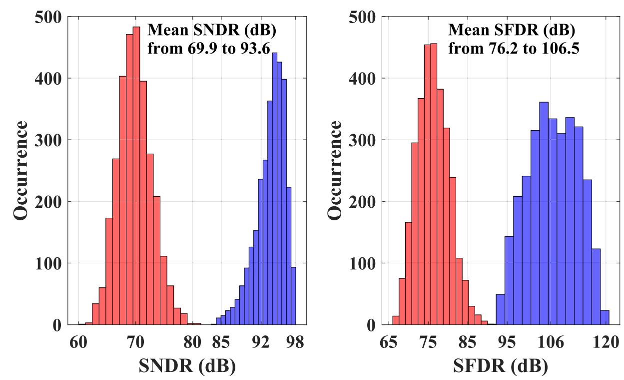

Fig. 7. MC simulation before/after the proposed calibration (3000 runs).

The results of the Monte Carlo simulation (3000 runs) before/after calibration are depicted in Fig. 7. The proposed calibration techniques enhance the mean values of SNDR and SFDR of the ADC by 23.7 and 30.3 dB, respectively, thereby demonstrating its robustness.

Fig. 8 shows the DNL and INL before and after calibration. After using the proposed calibration method, DNL is improved from −1 LSB/+3.56 LSB to −0.37 LSB/0.69 LSB, and INL is improved from −7.75 LSB/+7.66 LSB to −0.56 LSB/+0.49 LSB.

III. Circuits Implementation

A. High Linearity Sampling With Precharged Bootstrapped Switch

The sampling block plays a crucial role in determining the linearity of the ADC. The bootstrapped switch, which offers superior linearity compared with the CMOS switch, is widely utilized in high-precision ADC sampling modules.

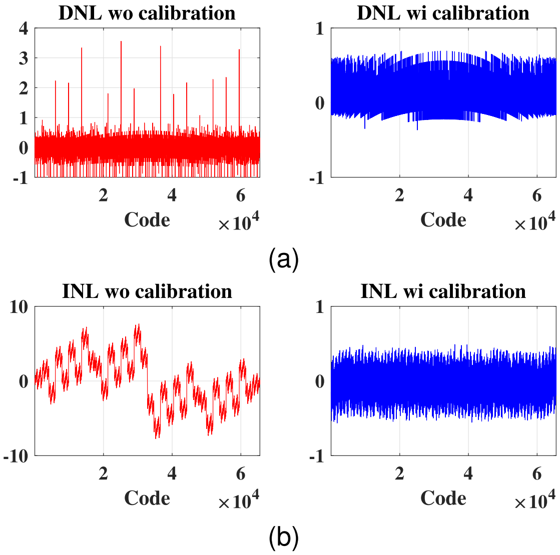

Fig. 8. DNL/INL before/after the proposed calibration.

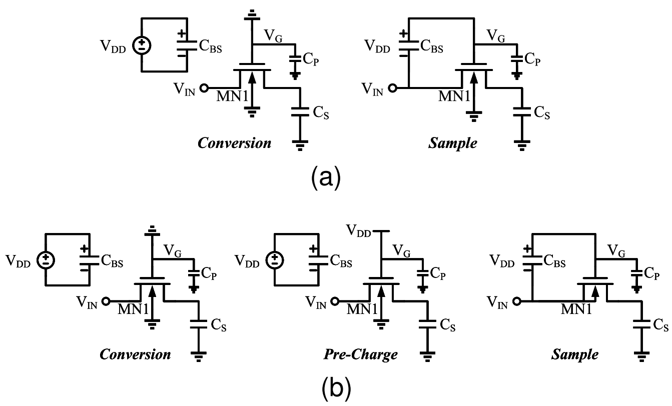

Fig. 9. Bootstrapped sampling circuits. (a) Conventional bootstrapped sampling block. (b) Proposed bootstrapped sampling block.

Fig. 9(a) shows the sampling block based on the bootstrapped switch in [23]. It operates in two phases: during the ADC conversion phase, the C_{BS} is charged to the voltage V_{DD}; during the sampling phase, the input signal V_{IN} is connected to the gate of the sampling transistor through C_{BS}, conducting the sampling nMOS (MN1). Ideally, the V_G equals the sum of V_{DD} and V_{IN}, so the MN1 has a constant V_{GS}, ensuring constant ON-resistance and preserving linearity in sampled signals.

In order to meet the bandwidth requirements, MN1 is designed with large dimensions and V_G is connected to many nodes, resulting in a significant parasitic capacitance C_P at this node. Considering C_P, V_G in the sampling phase can be expressed as follows:

$$V_G = \frac{C_{\mathrm{BS}}}{C_P + C_{\mathrm{BS}}}(V_{\mathrm{DD}} + V_{\mathrm{IN}}). \tag{21}$$

It can be seen that C_P not only increases the ON-resistance of the MN1 but also makes the ON-resistance related to the input signal, which will worsen the sampling linearity [24]. Increasing the size of C_{BS} or the MN1 can mitigate this effect, but it comes at the cost of larger area and power consumption overhead. The effect of increasing the size of the sampling transistor is also limited, as this not only increases C_P but also increases the parasitic capacitance of the drain–source junction, leading to extra nonlinearity. Moreover, self-biasing becomes impracticable during the conversion phase as it cannot determine the relationship between V_{IN} and V_{OUT}, leading to charge leakage between the substrate and active region. Therefore, the ON-resistance is also affected by the biasing effect, which also makes the ON-resistance related to the input signal. Considering these nonideal factors, the ON-resistance of the MN1 can be expressed as follows:

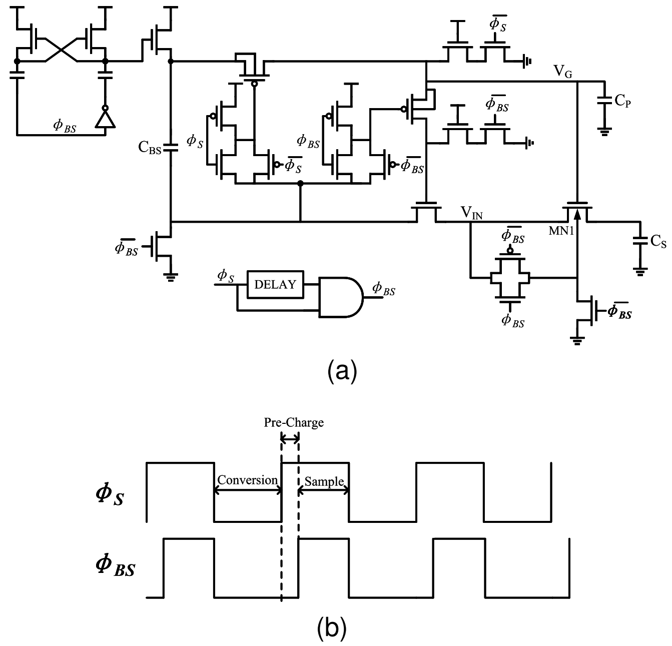

Fig. 10. Proposed precharged bootstrapped sampling circuits. (a) Circuit structure. (b) Clock phase.

$$R_{\mathrm{ON}} = \frac{K}{\frac{C_{\mathrm{BS}}}{C_{\mathrm{BS}}+C_P}V_{\mathrm{DD}} - \frac{C_P}{C_{\mathrm{BS}}+C_P}V_{\mathrm{IN}} - V_{\mathrm{TH}}(V_{\mathrm{IN}})} \tag{22}$$

where K is related to the size of MN1. It is worth noting that the trends of the parasitic capacitance and biasing effect on the ON-resistance are consistent, further worsening the sampling linearity.

Fig. 9(b) shows the proposed bootstrapped sampling block. In comparison with Fig. 9(a), a precharge phase has been introduced after the conversion phase to charge C_P to V_{DD}. Furthermore, during the sampling phase, the substrate of the MN1 is connected to the input signal to eliminate the body effect. The circuit implementation details are illustrated in Fig. 10. The proposed precharge phase can be generated using simple combinational logic of the sampling clock ϕ_S. After the conversion phase, the charge pump initially precharges C_P–V_{DD} and then V_{IN} crosses C_{BS} to fully activate the sampling switch during the sampling phase. In addition, the substrate of MN1 is self-biased for improved linearity by utilizing a complementary topology and connected to V_{SS} during the conversion phase for prevention of charge leakage. The ON-resistance of MN1 in this proposed sampling module can be represented as follows:

$$R_{\mathrm{ON}} = \frac{C_{\mathrm{BS}}}{C_P + C_{\mathrm{BS}}} * \frac{K}{\frac{C_{\mathrm{BS}}}{C_{\mathrm{BS}}+C_P}V_{\mathrm{DD}} - \frac{C_P * C_{\mathrm{BS}}}{(C_{\mathrm{BS}}+C_P)^2}V_{\mathrm{IN}} - V_{\mathrm{TH}}}. \tag{23}$$

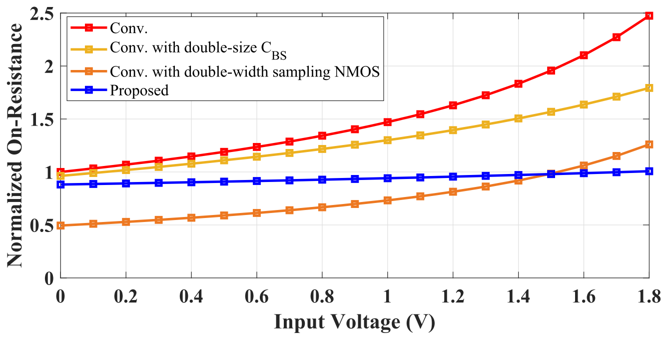

Fig. 11. Simulation for the proposed bootstrapped sampling circuit.

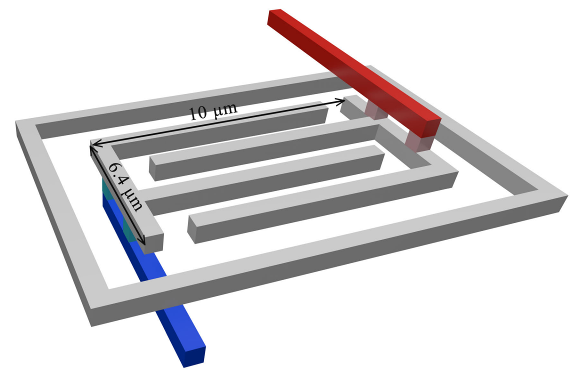

Fig. 12. Proposed custom-designed sub-femtofarad capacitor.

Although R_{ON} is still influenced by the input signal, the value of the ON-resistance and its dependence on the input signal are greatly reduced compared with (22).

The simulation results shown in Fig. 11 illustrate the normalized ON-resistance of both the conventional and proposed sampling blocks, which are based on the same size of C_{BS} and sampling switch MN1. In addition, Fig. 11 also depicts the normalized ON-resistance of doubling the capacitance of C_{BS} or the width of the sampling switch in the conventional bootstrapped sampling structure. It can be inferred that the ON-resistance of the sampling switch in [23] changes by approximately 147% due to parasitic capacitance and body effect. Increasing the size of C_{BS} by a factor of two reduces the variation in ON-resistance, but it remains at 86%, while comes with a doubling of the area and power consumption overhead. On the other hand, doubling the width of the sampling switch results in a halving of the ON-resistance, but also leads to a larger variation of 155% due to higher parasitic capacitance. In comparison, the proposed precharged bootstrapped sampling circuit not only has a 12% lower ON-resistance than [23] but also exhibits lower dependency on the input signal, with only a 14% change in resistance. The simulation demonstrates that, under identical conditions, the proposed sampling circuit exhibits higher linearity and wider sampling bandwidth. In other words, it achieves higher area and power efficiency.

B. Custom-Designed Sub-Femtofarad Capacitor

Due to the utilization of a binary-weighted CDAC without bridge, a unit capacitance of 0.5 fF is necessitated in this design. The PDK for the 180-nm process technology is unable to support metal–oxide–metal (MOM) or metal–insulator–metal (MIM) capacitors with such low capacitance values. Therefore, we propose a custom-designed capacitor which is illustrated in Fig. 12. There are two considerations when designing such a small capacitor: matching and accuracy of the capacitance.

The proposed scheme belongs to a lateral-field fringing capacitor constructed using interdigitated fingers, similar structures have been mentioned in [25] and [26], where the capacitance is primarily determined by the effective metal overlap length L_{eff}, finger spacing D, and metal height H, which is 0.4 μm in this process. It can be expressed as follows:

$$C \propto \frac{L_{\mathrm{eff}} \cdot H}{D}. \tag{24}$$

During calibration, the last 4 bits are considered to have an ideal weight and are used to calibrate the high-bit capacitance. Therefore, it is necessary to ensure that the maximum DNL of the last 4 bits of the CDAC is less than 0.5 LSB

$$\begin{aligned} 3\sigma_{\mathrm{DNL,max}} &= 3\frac{\sqrt{2^4-1}\sigma_{C_u}}{C_u} < 0.5 \\ &\Rightarrow \frac{\sigma_{C_u}}{C_u} < 4.3\%. \end{aligned} \tag{25}$$

According to [26], the mismatch of MOM capacitors is determined by the area of the capacitor, which is $L_{\mathrm{eff}} \cdot H$, and the mismatch is inversely proportional to the effective area and follows Pelgrom's inverse-area relation: when the area increases by k times, the capacitor mismatch becomes $1/\sqrt{k}$ times the original. The effective area of 1 $\mu\mathrm{m}^2$ of capacitance at this process node corresponds to a mismatch of 3.45%. Therefore, it can be inferred that the proposed customized capacitance exhibits a mismatch of 1.47%. Since eight groups of LSB capacitors are used for calibration, the equivalent unit capacitance mismatch is reduced to 0.52%.

The accuracy of custom-designed capacitors is influenced by the presence of parasitic capacitance. We define the two plates of the capacitor, with one connected to the comparator input as the top plate and the other end connected to the CDAC switches as the bottom plate. During the ADC conversion phase, where the top plate acts as a high-impedance node and the bottom plate switches between V_{REFP} and V_{REFN}, any parasitic capacitance between the top and bottom plates will impact the capacitance value.

The design incorporates isolation and routing measures to protect the high-impedance node of the CDAC. First, a shielding ring made from the same metal layer as the capacitor is used to prevent coupling between adjacent customized capacitor in the CDAC array. This ring is connected to V_{REFN}. In addition, the bottom plate is interconnected using a lower metal layer, while the top plate is linked through a higher metal layer to prevent coupling between these two plates.

Perturbations in the substrate may be coupled with the CDAC through the parasitic capacitance. In this design, the entire CDAC capacitor array is placed within a deep n-well (DNW), with PSUB within this DNW also being connected to V_{REFN}, such that substrate perturbations have the same effect

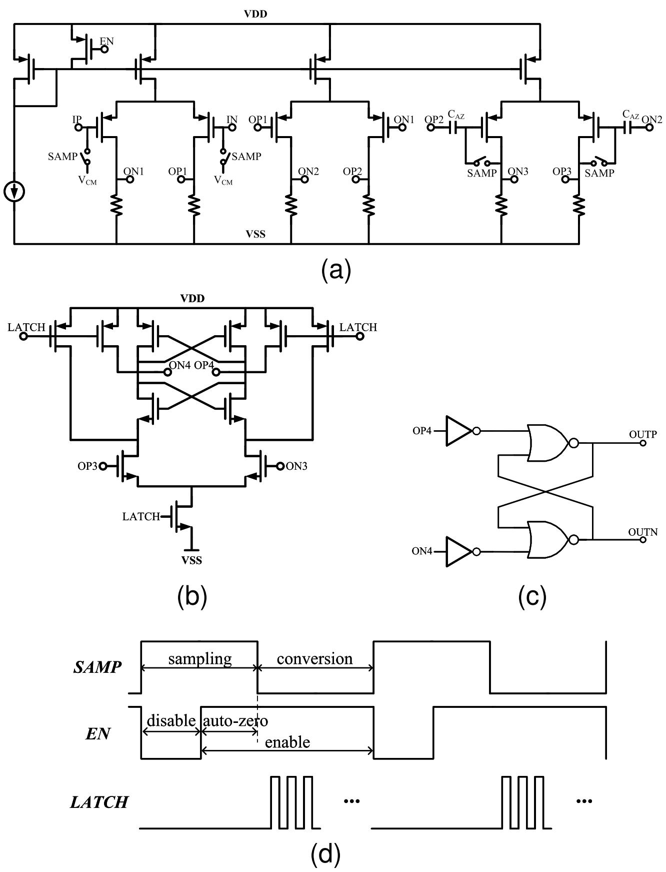

Fig. 13. Comparator circuits. (a) Pregain and autozero stages. (b) StrongARM latch. (c) SR latch. (d) Operation timing.

on CDAC-P and CDAC-N, and therefore do not affect the decision-making process of the comparator.

C. Comparator

The comparator used in the ADC utilizes a combination of pregain stages and a dynamic comparator to achieve high resolution and rapid output of comparison results, as shown in Fig. 13. The three stages of pregain, as depicted in Fig. 13(a), employ an identical structure with resistors serving as loads, providing a gain of approximately 60 dB. This ensures that the dynamic comparator can still reach the comparison threshold even when the differential input is as low as 0.5 LSB. The output common mode voltage is determined by the resistance and static current of each gain stage, eliminating the need for a common mode feedback circuit. The dynamic comparator shown in Fig. 13(b) adopts the StrongARM structure proposed in [27], [28], and [29], which exhibits zero static power consumption and direct generation of rail-to-rail output. The comparison result are stored using an SR latch, as depicted in Fig. 13(c), with its outcome being updated on every rising edge of the latch clock.

In order to meet the bandwidth and noise requirements during the conversion phase, the pregain stages exhibit a substantial static power consumption, which constitutes a significant portion of the overall power consumption of the ADC. Furthermore, even when the comparator is inactive during the sampling phase, the pregain stages still consume power, leading to significant power wastage in ADCs with a

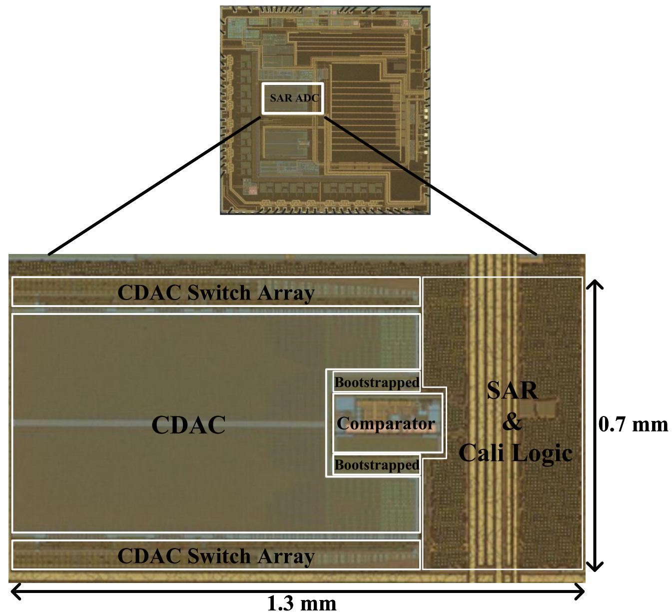

Fig. 14. Chip micrograph.

high sampling time ratio. In this design, the pregain stages are controlled by an enable signal EN, as shown in Fig. 13(d). When the ADC enters the sampling phase, EN is pulled down, and the pregain stages have no quiescent current, and after half of the sampling time has elapsed, EN is pulled up, then the pregain stages are enabled while entering the autozero phase, which results in a reduction in the comparator power consumption of about 25%.

Autozero is used to minimize offset and $1/f$ noise, which multiplexes the sampling clock as its control signal in this design. The comparator inputs are differentially shorted during the autozero phase, and the offset and $1/f$ noise of the pregain stages are stored on the C_{AZ} and then subtracted during conversion. The offset of the dynamic comparator is negligible because it is attenuated by a factor of the gain of the pre-gain stages.

The simulation demonstrates that the proposed comparator has an input-referred rms noise of 15 μV and a standard deviation of input-referred offset of 50 μV.

IV. Measurement Results

The prototype 16-bit SAR ADC chip was fabricated using a 1.8-V 180-nm process. Fig. 14 shows the die micrograph and the ADC core has dimensions of 1.3 $\times$ 0.7 mm. All measurement results presented in this section were obtained under room temperature conditions.

Fig. 15 shows the measured static performance. Before the calibration, the nominal weights of CDAC are used, exhibiting the maximum values of DNL and INL are +5.9/−1.0 LSB and +35.4/−34.3 LSB, respectively. After the proposed calibration, the weights are reconstructed thus eliminating the mismatch, the maximum values of DNL and INL are improved to +0.86/−0.85 LSB and +1.6/−1.2 LSB, respectively. Fig. 16 shows the measured ADC output spectrum before and after calibration at 100-kS/s sampling rate with a 97.5% full scale 1-kHz sine wave. The measured results are matched with

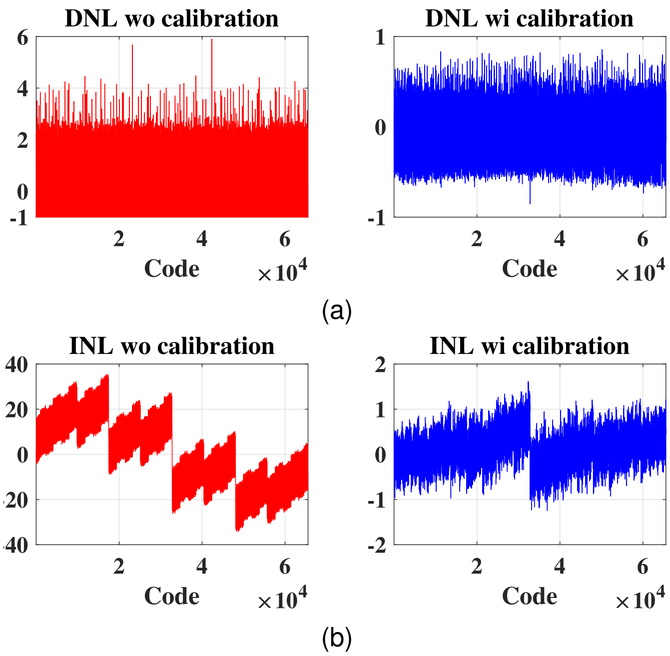

Fig. 15. Measured static performance. (a) DNL before/after calibration. (b) INL before/after calibration.

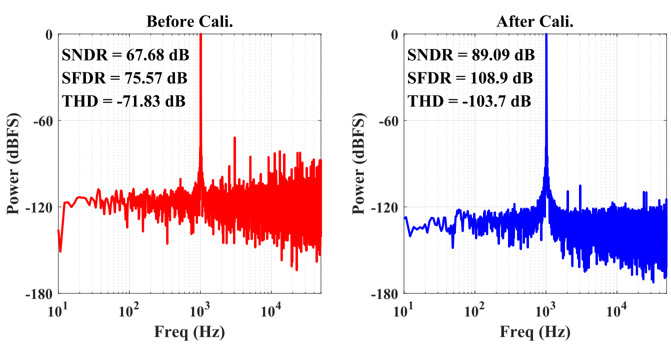

Fig. 16. ADC output spectrum before/after calibration at a sampling frequency of 100 kS/s with 1-kHz input.

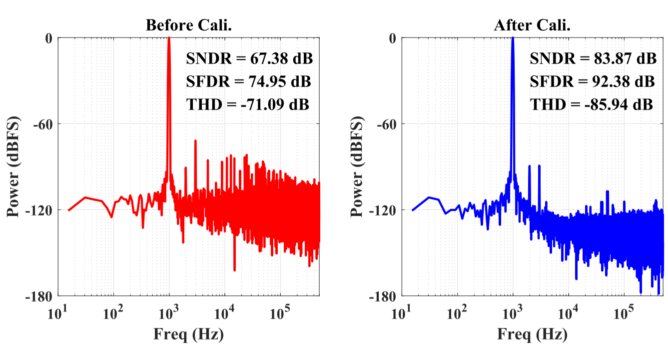

Fig. 17. ADC output spectrum before/after calibration at a sampling frequency of 1 MS/s with 1-kHz input.

the behavioral simulation proposed in Section III. At a low sampling rate of 100 kS/s, the linearity of the ADC is primarily constrained by capacitor mismatch. The measurement reveals a significant improvement of 21.4/33.3 dB in SNDR/SFDR after calibration, demonstrating the effectiveness of the proposed mismatch calibration method. Fig. 17 shows the spectrum when ADC works at 1 MS/s. After the mismatch calibration,

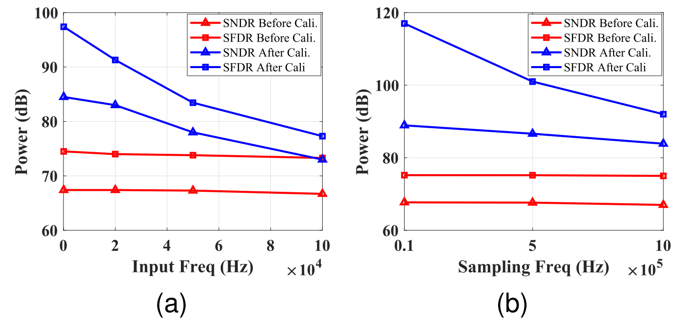

Fig. 18. Measured ac performance. (a) SNDR/SFDR versus input frequency at a sampling of 1 MS/s. (b) SNDR/SFDR versus sampling frequency at 1-kHz input frequency.

TABLE I
PERFORMANCE COMPARISON

	This work	Shen [19]	Li [6]	Bodnar [7]	Chen [8]
Type	SAR	Flash-SAR	Pipe-SAR	Pipe-SAR	Pipe-SAR
Process [nm]	180	55	180	40	22
Supply [V]	1.8	3.3/1.2	3.3/1.8	3.3	1
Resolution [bits]	16	16	22	20	13
Speed [MS/s]	1	16	12	40	475
Calibration	On Chip	On Chip	Off Chip	Off Chip	On Chip
DNL/INL [LSB]	0.86/1.6	0.8/2.3	NA	0.28/2.3	0.51/1.22
SNDR /SFDR [dB]	84/92	78/98	91.3/99.2	93.7/119	65.9/73
Power [mW]	6.7	16.3	9.43	50	9.93
Area [mm^2]	0.91	0.55	0.6	0.9	0.038
FoM_S [dB]	162.7	165	179.3	179.7	169.7

the ADC achieves SNDR and SFDR of 83.87 and 92.38 dB, respectively, which is an improvement of 16.49 and 17.43 dB compared with the noncalibrated case. The reduced SFDR observed in comparison with Fig. 16 is attributed to the coupling between the chip clock (approximately 32 times the ADC sampling frequency) and the ADC negative input during chip operation in the ADC test mode. Fig. 18(a) presents the dynamic performance of the ADC versus the input frequency at a sampling rate of 1 MS/s. At frequencies below 10 kHz which is the typical application of this chip, the calibration provides around 17.6-dB SNDR and 20-dB SFDR improvement. Fig. 18(b) shows the dynamic performance of the ADC versus the sampling frequency, using a 1-kHz sinewave input. The SNDR and SFDR are above 83 and 92 dB, respectively, up to a sampling frequency of 1 MS/s. Notably, the calibrated ADC achieves an impressive SFDR of 117 dB at a sampling rate of 10 kS/s, surpassing the uncalibrated case by 42 dB. Under 1.8-V analog and digital supply with the ADC running at 1 MS/s, the total power dissipation is 6.745 mW, comprising 6.34 mW comparator power, 268.3 μW logic power, and 131.7 μW consumed by the DAC switching. Unlike the ADC using background calibration, the digital power dissipation of the proposed ADC is negligible because the calibration only executes once before the ADC normal operation. Comparison with several high-resolution SAR ADCs using capacitor mismatch calibration is given in Table I.

V. CONCLUSION

A 16-bit 1-MS/s SAR ADC fabricated in a 1.8-V 180-nm CMOS process is presented in this article. Capacitor mismatch in CDAC is addressed by using proposed self-calibration techniques, including floating operation and LSB capacitors round-robin. The floating capacitor operation eliminates the impact of static offset, thus preventing the sub-ADC from saturating during the calibration process. The random mismatch of the LSB capacitors themselves limits the calibration accuracy, which is addressed in this article by using the LSB capacitor's round-robin. The precharged bootstrapped sampling switch not only achieves a lower ON-resistance but also reduces the dependence of ON-resistance on the input signal. The binary-weighted CDAC without a bridge is facilitated by the implementation of a custom-designed 0.5-fF lateral-field fringing capacitor with anti-interference techniques. The comparator employs a combination of pre-gain stages and a dynamic comparator, integrated with autozero technique to minimize static offset and $1/f$ noise. Behavioral simulations and measurement results demonstrate the effectiveness of the proposed techniques.

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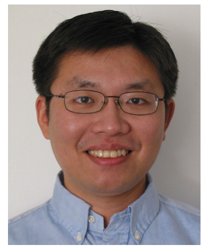

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